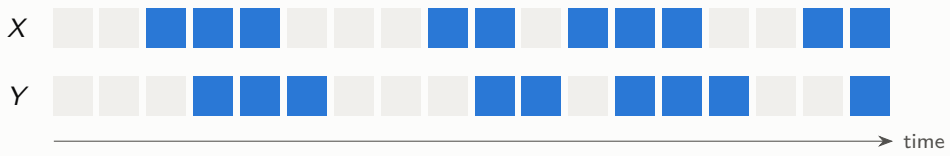
# Do information-theoretic causality measures agree?

## Transfer entropy, information flow and causal strength on tiny systems we can solve exactly

Michael Rowlands

## A puzzle

Two binary signals, recorded once per time step (filled = 1):




$Y$ looks exactly like $X$, one step later. So…

- Does $X$ *drive* $Y$?
- Or does some third thing drive both, reaching $X$ first?
- And how would we put a *number* on "how much $X$ influences $Y$"?

Scientists ask this about brain regions, climate indices and stock prices. A popular tool is **transfer entropy**, often described as "information flowing from $X$ to $Y$". Today: when does that description hold, and when does it fail?

# The plan

**Three measures** of "how much $X$ influences $Y$"

1. Transfer entropy (Schreiber, 2000)
2. Information flow (Ay & Polani, 2008)
3. Causal strength (Janzing et al., 2013)

**Three toy systems** built so they disagree

1. a hidden common driver
2. an XOR ("it takes two")
3. a slowly changing signal that is copied

**Why toy systems?**

Every variable is a single bit, so each system has at most $2^5 = 32$ possible outcomes. We can write down the *whole* probability table and compute every measure **exactly**.

No data, no sampling noise: if two measures disagree, the disagreement is about what they *mean*, not about statistics.

First we need two toolkits: measuring information in bits, and talking precisely about causes.

## Where we are

- Information in bits
- Causes and interventions
- Three measures of influence
- Three systems where they disagree
- Why they disagree
- Real data, and takeaways

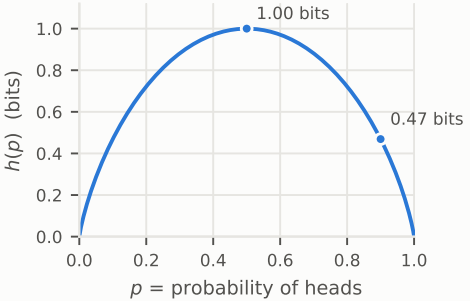
## Entropy: uncertainty measured in bits

For a random variable $X$ with probabilities $p(x)$:

$$H(X) = -\sum_{x} p(x)\, \log_2 p(x)$$

- The average number of yes/no questions needed to guess $X$.
- Fair coin: $H = 1$ bit.
- Coin with $P(\text{heads}) = 0.9$: $H \approx 0.47$ bits. It is easier to guess.
- A coin that always lands heads: $H = 0$.

For a biased coin we write $h(p)$, the *binary entropy*.



# Mutual information: what one variable tells you about another

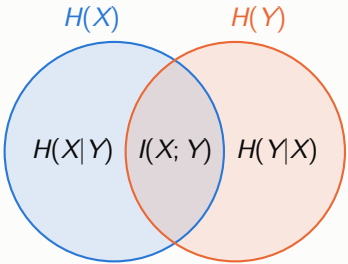
$$I(X;Y) = H(Y) - H(Y \mid X)$$

The bits of uncertainty about $Y$ that knowing $X$ removes.

- Symmetric: $I(X;Y) = I(Y;X)$.
- $I(X;Y) = 0$ exactly when $X$ and $Y$ are independent.

**Examples** with a fair coin $X$:

- $Y = X$: $I = 1$ bit
- $Y$ an unrelated coin: $I = 0$
- $Y = X$, but flipped with probability $\varepsilon$: $I = 1 - h(\varepsilon)$



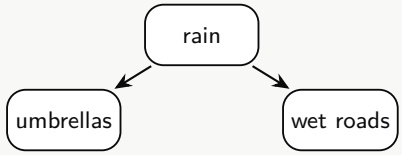

The overlap is the information $X$ and $Y$ share.

# Conditional mutual information: what's new, given what you know

$$I(X; Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z)$$

What $X$ tells you about $Y$ *once you already know* $Z$. Conditioning can move it either way:

**It can remove dependence**




Umbrellas predict wet roads: $I(X; Y) > 0$. Once you know whether it rained, umbrellas add nothing: $I(X; Y \mid Z) = 0$.

**It can create dependence**

$Y = X \oplus Z$, the XOR of two fair coins:

| $X$ | $Z$ | $Y$ | $X$ | $Z$ | $Y$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 |

Alone, $X$ says nothing about $Y$: $I(X; Y) = 0$. Given $Z$, $X$ decides $Y$: $I(X; Y \mid Z) = 1$ bit.

## Relative entropy: how different are two distributions?

$$D(P \,\|\, Q) = \sum_x P(x)\, \log_2 \frac{P(x)}{Q(x)}$$

Also called the *Kullback–Leibler divergence*.

- The extra bits you pay, on average, if the world follows $P$ but you planned for $Q$.
- Always $\geq 0$, and $= 0$ exactly when $P = Q$.
- Not symmetric, so not a true distance.

Example: $P =$ fair coin, $Q = 90/10$ coin gives $D(P\|Q) \approx 0.74$ bits.

**Mutual information is a relative entropy**

$$I(X;Y) = D\big(P(x,y) \,\|\, P(x)\,P(y)\big)$$

How far the real joint distribution is from "$X$ and $Y$ independent".

We'll use this idea again: one causal measure compares the real world with a world where a link is cut.

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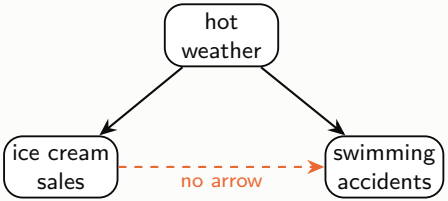
# Correlation is not causation

Ice cream sales and swimming accidents rise and fall together. Does ice cream cause accidents?

No: hot weather drives both. It is a **common cause**, or **confounder**.

A **causal graph** draws this:

- one node per variable,
- an arrow $A \to B$ when $A$ is a *direct input* to the mechanism that produces $B$.



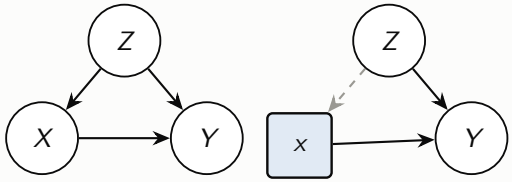

Ice cream *predicts* accidents, but doesn't *cause* them.

# Seeing versus doing

**Seeing**: $P(Y \mid X = x)$
Among the days when $X$ *happened* to be $x$, how is $Y$ distributed?

**Doing**: $P(Y \mid \mathrm{do}(X = x))$
Reach in and *force* $X = x$, as in an experiment, overriding whatever normally sets $X$.

In the graph, forcing $X$ means **deleting the arrows into** $X$.




seeing: as it runs

doing: $X$ set by hand

**Ice cream again**

There is no arrow from ice cream to accidents, so handing out free ice cream changes nothing: $P(\text{acc.} \mid \mathrm{do}(\text{ice cream})) = P(\text{acc.})$. Seeing mixes in whatever travels with $X$; doing isolates $X$ itself.

## Computing $\mathrm{do}(\cdot)$ when you know the graph

A causal model is a graph plus one **mechanism** per variable, $P(v \mid \text{parents of } v)$:

$$P(v_1, \ldots, v_n) = \prod_i P(v_i \mid \mathrm{pa}_i)$$

To compute $\mathrm{do}(X = x)$, delete $X$'s mechanism, set $X = x$, and keep every other mechanism:

$$P(v \mid \mathrm{do}(X = x)) = \prod_{i \neq X} P(v_i \mid \mathrm{pa}_i)\Big|_{X=x}$$

For the graph on the previous slide:

$$P(y \mid \mathrm{do}(x)) = \textstyle\sum_z P(z)\, P(y \mid x, z)$$
$$P(y \mid x) = \textstyle\sum_z P(z \mid x)\, P(y \mid x, z)$$

They differ exactly when $Z$ travels with $X$.

**The catch**

You need the *graph*. The same data can fit several graphs, and data alone can't always tell you which way the arrows point.

# Time series as causal graphs

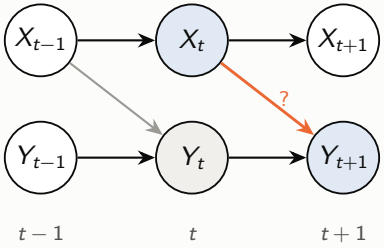
Unroll time: every variable at every step gets its own node. Causes come before their effects.

**The question for the rest of the talk:**

How much does $X_t$ influence $Y_{t+1}$?

$Y_t$ is the target's *own past*. Keep an eye on it.

All variables are bits, and each process has been running long enough to be stationary: its statistics don't change over time.



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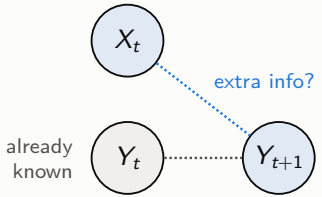
## Measure 1: transfer entropy

$$\mathrm{TE}(X \to Y) = I(Y_{t+1};\, X_t \mid Y_t)$$

"Does $X$'s present help predict $Y$'s next value, *beyond* what $Y$'s own past already tells us?"

$$= H(Y_{t+1} \mid Y_t) - H(Y_{t+1} \mid Y_t, X_t)$$

- ✓ Needs only observed data: no graph, no experiments.
- ✓ Widely used, and often read as "information flow from $X$ to $Y$".
- ✗ Purely *predictive*. It never asks what would happen under an intervention.




Schreiber, *Phys. Rev. Lett.* 2000.

## Measure 2: information flow (Ay & Polani)

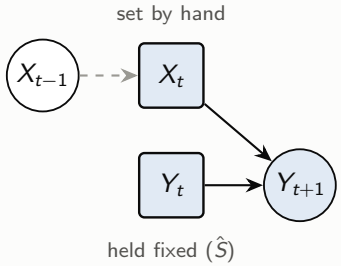
A thought experiment: **set $X_t$ by hand**, at random with its usual frequencies, and watch $Y_{t+1}$.

$I(X_t \to Y_{t+1})$ is the mutual information between the value you set and $Y_{t+1}$.

**Imposing a set** $S$ (the hat means "set by hand"): also fix $S$ by intervention, then ask how much $X_t$ still matters. This gives $I(X_t \to Y_{t+1} \mid \hat{S})$. We impose $S = \{Y_t\}$, plus any other input to $Y_{t+1}$.

- ✓ Genuinely causal: it measures the response to interventions.
- ✗ Needs the causal graph, to compute $\mathrm{do}(\cdot)$.

Ay & Polani, *Adv. Complex Syst.* 2008.




**Formally**

$\hat{p}(s, x, y) = p(s)\ p(x \mid \mathrm{do}(s))\ p(y \mid \mathrm{do}(s, x))$

flow $= I_{\hat{p}}(X; Y \mid S)$, an ordinary conditional MI.

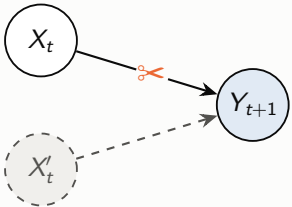
# Measure 3: causal strength (Janzing et al.)

Think of the arrow $X_t \to Y_{t+1}$ as a **wire**. **Cut it**, and feed $Y$'s input with an independent random copy $X'_t$ that has the same statistics as $X_t$.

$$C_{X \to Y} = D\big(P \,\|\, P_{\text{cut}}\big)$$

The relative entropy between the real world and the world with the wire cut.

- $C = 0$ exactly when cutting changes nothing, e.g. when there is no wire.
- For a lone arrow $X \to Y$, $C = I(X; Y)$.
- ✗ Also needs the causal graph.




independent copy,
same distribution as $X_t$

Janzing, Balduzzi, Grosse-Wentrup & Schölkopf, *Ann. Stat.* 2013.

## Four numbers for one question

| | quantity | needs | question it answers |
|---|---|---|---|
| Lagged MI | $I(X_t; Y_{t+1})$ | data | Do $X$ now and $Y$ next vary together? |
| Transfer entropy | $I(Y_{t+1}; X_t \mid Y_t)$ | data | Does $X$ help predict $Y$ beyond $Y$'s past? |
| Ay–Polani flow | $I(X_t \to Y_{t+1} \mid \hat{S})$ | graph | If I *set* $X$, does $Y$ respond? |
| Janzing strength | $D(P \,\|\, P_{\text{cut}})$ | graph | How much does cutting $X \to Y$ change things? |

Lagged MI is a baseline: plain correlation, in bits. We also report Ay–Polani flow with nothing imposed, written $I(X_t \to Y_{t+1})$.

All values below are exact, in bits, computed in Python with the `dit` library.

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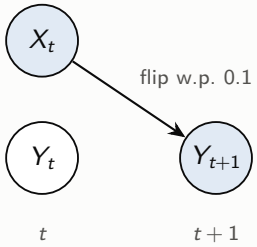
## First, a sanity check: a plain noisy wire

$Y_{t+1}$ copies $X_t$, but flips it with probability 0.1. $X$ is a fresh fair coin every step, and $Y$ has no memory.

All five quantities (MI, TE, both Ay–Polani flows, Janzing) give the same answer:

$$1 - h(0.1) = 0.531 \text{ bits}$$

When nothing tricky is going on, **all the measures agree**. Now let's make things tricky.



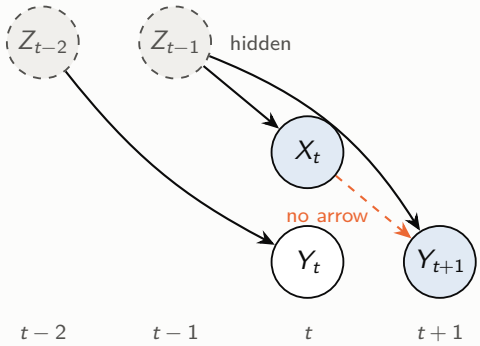
# System 1: a hidden driver with staggered delays

A hidden coin $Z$ is flipped every step.

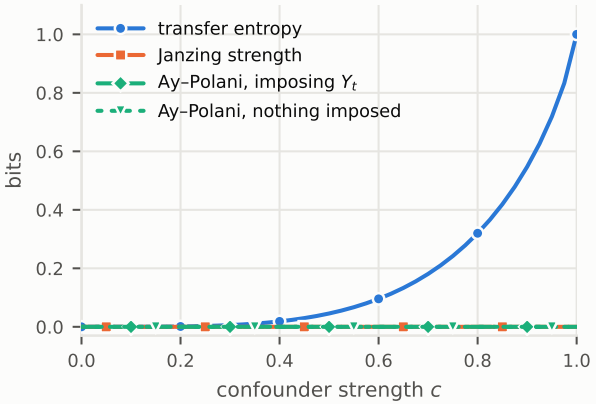
- $X$ copies $Z$ after **one** step.
- $Y$ copies $Z$ after **two** steps.
- Each copy is faithful with strength $c$ (correlation $c$; $c = 1$ means a perfect copy).

There is **no arrow** from $X$ to $Y$. But $X$ sees $Z$ one step before $Y$ does, so $X$ *predicts* $Y$'s next value.

This is the ice-cream problem, stretched out in time.



# Hidden driver: transfer entropy sees a flow that isn't there




- **Transfer entropy** rises to 1 bit as $c \to 1$:

$$\mathrm{TE} = 1 - h\left(\frac{1-c^2}{2}\right)$$

  It reads 0.55 bits at $c = 0.9$.
- **Both causal measures** are exactly 0 for every $c$. Setting $X_t$ by hand does nothing to $Y_{t+1}$, and there is no wire to cut.

TE answers "does $X$ help *predict* $Y$?". Here the answer is yes, even though $X$ has no causal effect on $Y$.

# The catch: causal measures need the right graph

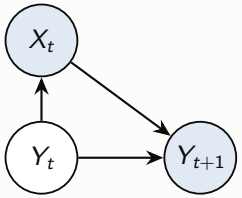
Suppose we can't observe $Z$. From $X$ and $Y$ alone, the natural graph to draw is the one on the right.

Run both causal measures on the same data with *this* graph:

| | true graph | analyst's graph |
|---|---|---|
| TE | 0.547 | 0.547 |
| Ay–Polani | 0 | 0.547 |
| Janzing | 0 | 0.547 |

With the wrong graph they make **exactly the same mistake as TE**.

analyst's graph ($Z$ unseen)




**Lesson**

The causal measures don't win because of a cleverer formula. They win because we *handed them the truth* about the graph.

## System 2: XOR, where it takes two

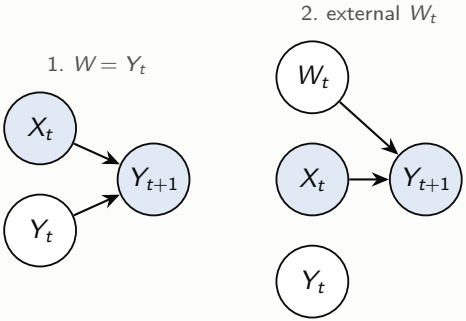
$$Y_{t+1} = X_t \oplus W$$

with $X_t$ and $W$ independent fair coins. Knowing only $X$ tells you *nothing* about $Y$; knowing both tells you *everything*.

Two versions, which differ only in **who the partner $W$ is**:

1. $W = Y_t$: the partner is $Y$'s own past.
2. $W = W_t$: the partner is some other signal, and $Y_t$ is irrelevant.

Version 1 is the example of James, Barnett & Crutchfield (2016).



## XOR: the answer depends on what you condition on

| | MI | TE | AP | AP $\mid \hat{S}$ | Janzing |
|---|---|---|---|---|---|
| 1. $W = Y_t$ | 0 | 1 | 0 | 1 | 1 |
| 2. $W = W_t$ | 0 | 0 | 0 | 1 | 1 |

AP: flow with nothing imposed. $\hat{S} = \{Y_t\}$ in version 1 and $\{Y_t, W_t\}$ in version 2.
Shaded: says "no influence", although $X_t$ clearly influences $Y_{t+1}$.

- **Looking at $X$ alone** (MI, or flow with nothing imposed) always misses the XOR, because on average $X$ has no effect on $Y$.
- **Transfer entropy** gets 1 bit only if the partner *happens to be* $Y$'s past. With an external partner it sees nothing.
- **Imposing the partner**, or **cutting the wire**, reveals the full bit either way.

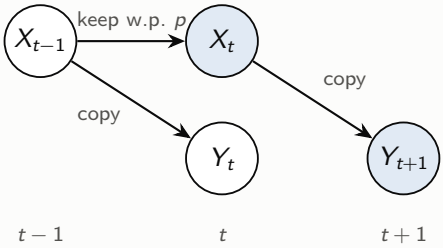
Adding noise $\varepsilon$ to the XOR turns every 1 above into $1 - h(\varepsilon)$ and leaves the zeros at 0.

# System 3: a slow signal, copied

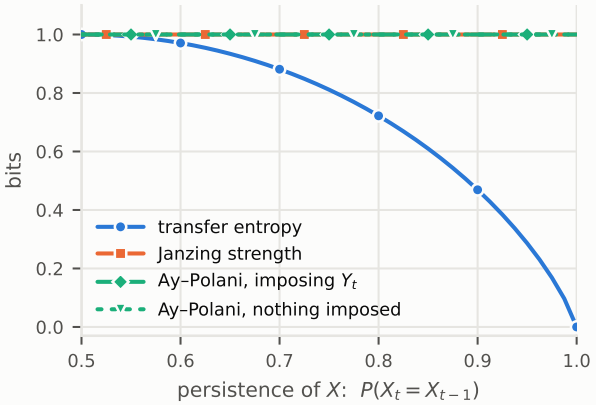
- $X$ switches slowly: it keeps its value with probability $p$, its *persistence*.
- $Y$ copies $X$ **perfectly**, one step later: $Y_{t+1} = X_t$.

This is the opening puzzle. $Y$ is completely driven by $X$, so the causal influence is a full bit.

But when $X$ rarely changes, $Y$'s own past $Y_t = X_{t-1}$ is already a great predictor of $X_t$, and so of $Y_{t+1}$.



# Synchronization: perfect coupling, vanishing transfer entropy




- **Causal measures**: 1 bit, whatever $p$ is. Setting $X_t$ fully determines $Y_{t+1}$.
- **Transfer entropy** $= h(p) \to 0$ as $X$ becomes predictable. At $p = 0.95$ it is 0.29 bits.

TE only counts information about $Y_{t+1}$ that is *new* relative to $Y$'s past. A perfectly coupled but predictable pair looks like almost no transfer.

At $p = 1$ this is the "synchronized chains" example of Ay & Polani: TE $= 0$, influence $= 1$ bit.

# All systems at a glance

| system | $X_t \to Y_{t+1}$? | MI | TE | AP | AP $\mid \hat{S}$ | Janzing |
|---|---|---|---|---|---|---|
| Noisy wire (control) | yes | 0.53 | 0.53 | 0.53 | 0.53 | 0.53 |
| Hidden driver, $c = 0.9$ | no | 0.55 | 0.55 | 0 | 0 | 0 |
| same data, analyst's graph | no | 0.55 | 0.55 | 0.55 | 0.55 | 0.55 |
| XOR, $W = Y_t$ | yes | 0 | $1^\dagger$ | 0 | 1 | 1 |
| XOR, external $W_t$ | yes | 0 | 0 | 0 | 1 | 1 |
| Slow copy, $p = 0.95$ | yes | 1 | 0.29 | 1 | 1 | 1 |

Shaded: misleading about whether, or how strongly, $X_t$ influences $Y_{t+1}$.
$^\dagger$ The right number for a subtle reason, as the next section shows.

Transfer entropy gives a **false positive** (hidden driver), a **false negative** (external XOR) and a large **underestimate** (slow copy).

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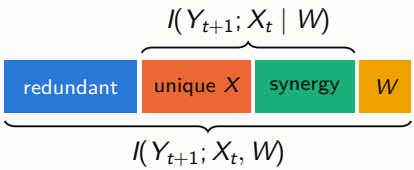
Why they disagree

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## Splitting information into kinds

Two sources, $X_t$ and a partner $W$, carry information about a target $Y_{t+1}$. *Partial information decomposition* (Williams & Beer, 2010) splits it into four parts:

- **Redundant**: either source alone would tell you.
- **Unique to** $X$: only $X$ tells you.
- **Unique to** $W$: only $W$ tells you.
- **Synergistic**: only the two *together* tell you, as in XOR.

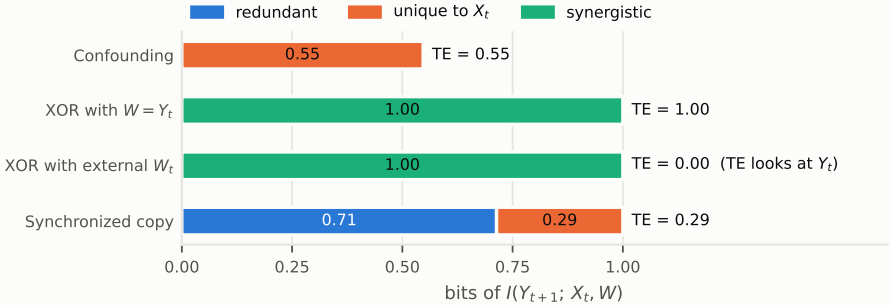



**The identity that explains transfer entropy**

With the partner $W = Y_t$: $\mathrm{TE} = I(Y_{t+1}; X_t \mid Y_t) =$ unique to $X +$ synergy

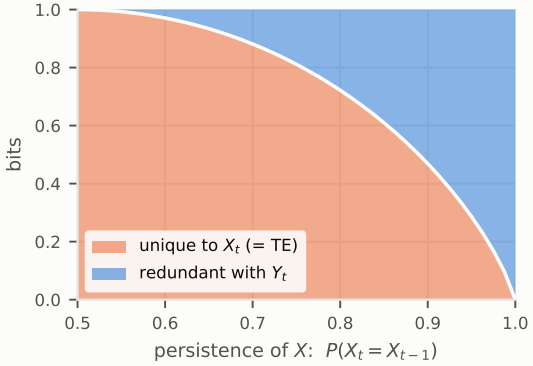
TE leaves out the redundant part, and counts synergy as if it belonged to $X$ alone.

# What transfer entropy is made of




- **XOR,** $W = Y_t$: the whole bit is *synergy*. TE credits it to $X$, although it exists only in the pair.
- **External XOR**: synergy between $X_t$ and $W_t$. TE's partner is $Y_t$, so it sees none of it.
- **Slow copy**: most of the bit is *redundant* with $Y$'s past, and TE drops it.
- **Hidden driver**: looks like clean *unique* information. The decomposition is observational too, so it can't see $Z$.

# Slow copy: redundancy crowds out transfer entropy




Together $X_t$ and $Y_t$ always know the full 1 bit about $Y_{t+1}$.

As $X$ becomes more persistent, more of that bit is also known to $Y$'s past, because $Y_t = X_{t-1}$ usually equals $X_t$. That part is redundant.

TE keeps only the **unique** part, $h(p)$. It measures *new* information, not coupling strength.

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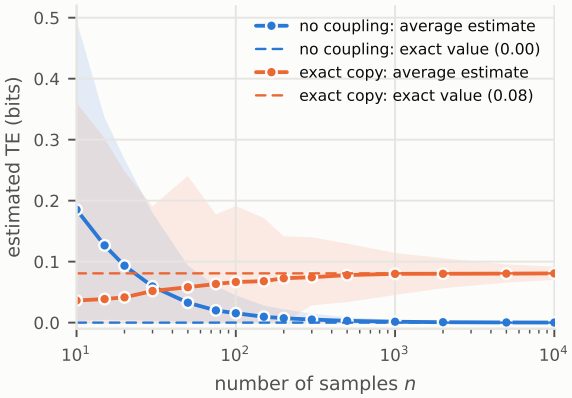
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# With finite data it gets worse




Shaded: middle 90% of 2,000 repeated experiments.

In practice we *estimate* TE: count how often each outcome occurs in $n$ samples, then plug the frequencies into the formula.

- With **no coupling at all**, the average estimate is 0.19 bits at $n = 10$ and 0.015 at $n = 100$. It shrinks only like $1/(n \ln 2)$.
- For a deterministic copy the estimate is biased *downward*.
- For $n \leq 30$ the uncoupled system looks *more* "causal" than the perfectly coupled one.

Always compare an estimate against a null, such as shuffled data.

## Takeaways

1. **Transfer entropy measures prediction, not causation.** It answers "does $X$ help forecast $Y$ beyond $Y$'s past?". That question has a different answer from "does $X$ influence $Y$?" when there is
   - a hidden common driver (false positive),
   - synergy with another input (misattributed, or missed entirely),
   - a predictable, synchronized source (underestimate).
2. **Interventional measures fix this, given the right causal graph.** With the wrong graph they inherit the same errors.
3. **Finite samples add spurious transfer entropy** of about $1/(n \ln 2)$ bits, so test against a null.
4. **Pick the measure that matches your question**: forecasting, or "what if I intervened?".

## Questions to think about

1. In the hidden-driver system, which extra variable would you condition on to make transfer entropy report 0? What does that suggest about using TE in practice?
2. In the slow-copy system, TE $\to 0$ as $X$ becomes predictable. Is that a bug or a feature? When would you *want* this behavior?
3. For the XOR, Ay–Polani flow is 0 with nothing imposed and 1 bit when imposing $W$. Which one is "the" causal influence of $X$? Do we need to choose?
4. Why is the plug-in estimate biased *upward* with no coupling, but *downward* for a deterministic copy? Hint: think about outcomes that never show up in a small sample.

## References and code


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- D. Janzing, D. Balduzzi, M. Grosse-Wentrup & B. Schölkopf (2013). Quantifying causal influences. *Ann. Stat.* 41, 2324.
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- P. L. Williams & R. D. Beer (2010). Nonnegative decomposition of multivariate information. arXiv:1004.2515.
- J. Pearl (2009). *Causality*, 2nd ed. Cambridge University Press. (seeing vs. doing)


**Code:** every number in these slides is computed exactly in the `causality-measures` repository (Python, `dit`). Run `uv run pytest`, then the scripts in `experiments/`.